

SHARP METRIC-DIMENSION RESULTS AROUND GEHÉR'S PROBLEM 1

AUTOMATED PROOF ATTEMPT FOR ARXIV:1507.05114

ABSTRACT. For a d -dimensional normed space, we prove that every finite resolving set has at least $d + 1$ points. Norms whose unit ball has a facet have no finite resolving set at all. At the opposite end, the metric dimension of ℓ_p^d is exactly $d + 1$ for every $1 < p < \infty$, with the standard simplex as an optimal resolving set. We also prove the same conclusion for a quantitative $C^{1,1}$ neighbourhood of Euclidean norms containing genuinely non-Hilbert examples. These results answer the literal non-strict question negatively and settle large strictly convex classes, but do not resolve the universal strictly convex version.

1. SOURCE QUESTION AND SCOPE

Let $(X, \|\cdot\|)$ be a finite-dimensional real normed space. A set $R \subset X$ *resolves* X if

$$(\|x - r\|)_{r \in R} = (\|y - r\|)_{r \in R} \implies x = y.$$

We write $\text{mdim } X$ for the least cardinality of a finite resolving set, and $\text{mdim } X = \infty$ if none exists.

Section 4, Problem 1 of Gehér [1] asks exactly:

“Is the metric dimension of every d -dimensional (strictly convex) normed space less than or equal to $d + 1$?”

The next paragraph says that it was not clear whether the metric dimension could be smaller than $d + 1$. The exact source context is reproduced below.

of complex inner product spaces of dimension at least three as well. However in that case, one should be aware that in the definition of (SRS d) affine independence is real affine independence, and also convex hull is real convex hull.

Next, we say that the metric dimension of X is $\delta \in \mathbb{N}$ if there exists a resolving set for X with δ elements, but there is no resolving set with less elements. The metric dimension of a d -dimensional Euclidean space is clearly $d + 1$. From our results it is clear that usually a set of $d + 1$ affine independent points is not a resolving set for a d -dimensional normed space. Therefore it is reasonable to ask the following question.

Problem 1. *Is the metric dimension of every d -dimensional (strictly convex) normed space less than or equal to $d + 1$?*

The reason why we have asked “less than or equal to” is that it is not clear whether the metric dimension of a d -dimensional normed space can be less than $d + 1$. However, it is quite plausible that such a phenomena cannot happen.

The parenthetical admits two natural readings: all normed spaces, or at least all strictly convex ones. The first is false by Theorem 2. The second is verified here for broad classes but remains unresolved in full.

2. PROOF INTUITION

There are three short mechanisms.

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- (1) Distances to m landmarks give a continuous coordinate map into $\mathbb{R}^m$. Invariance of domain forbids injectivity when $m \leq d$.
- (2) A facet creates an open cone on which the norm is exactly linear. Far inside that cone, a finite landmark set cannot detect motion tangent to the facet.
- (3) In ℓ_p^d , subtracting the equality at the origin from the equality at the i th coordinate landmark isolates one strictly monotone scalar function of the i th coordinate. Near a Hilbert norm, the same relative distance map is an invertible linear map plus a strict Lipschitz contraction.

3. A UNIVERSAL SHARP LOWER BOUND

Theorem 1 (No d landmarks). *Let X be a real normed space of dimension $d \geq 1$. If X has finite metric dimension, then*

$$\text{mdim } X \geq d + 1.$$

Proof. Identify X linearly and topologically with $\mathbb{R}^d$. Suppose that $R = \{r_1, \dots, r_m\}$ resolves X , where $m \leq d$, and define

$$F : X \longrightarrow \mathbb{R}^m, \quad F(x) = (\|x - r_1\|, \dots, \|x - r_m\|).$$

This map is continuous and injective. Pad it with $d - m$ zero coordinates to obtain a continuous injection $\tilde{F} : \mathbb{R}^d \rightarrow \mathbb{R}^d$. By Brouwer's invariance-of-domain theorem, $\tilde{F}(\mathbb{R}^d)$ is open.

If $m < d$, this image lies in a proper coordinate subspace and hence has empty interior, a contradiction. If $m = d$, then $F(X) \subset [0, \infty)^d$. Moreover, the first coordinate of $F(r_1)$ is zero, so no Euclidean neighbourhood of $F(r_1)$ is contained in the nonnegative orthant. Thus $F(X)$ is not open, again a contradiction. $\square$

This proves the lower bound conjectured in the paragraph following Problem 1, without any convexity or smoothness assumption.

4. FACETS FORCE INFINITE METRIC DIMENSION

Let B be the closed unit ball. By a *facet* we mean an exposed face of affine dimension $d - 1$.

Theorem 2 (Facet obstruction). *Let $d \geq 2$. If the unit ball of X has a facet, then $\text{mdim } X = \infty$.*

Proof. Choose $\varphi \in X^*$ such that $\varphi \leq 1$ on B and

$$F = B \cap \{\varphi = 1\}$$

is a facet. The cone

$$C = \{tz : t > 0, z \in \text{relint } F\}$$

is a nonempty open subset of X , and

$$(1) \quad \|z\| = \varphi(z) \quad (z \in C).$$

Let $R \subset X$ be finite. Fix $q \in C$ and a nonzero $w \in \ker \varphi$. Since C is open and conic, for all sufficiently large T and all $r \in R$ both

$$Tq - r \quad \text{and} \quad Tq + w - r$$

belong to C . Put $x = Tq$ and $y = Tq + w$. Then $x \neq y$, while (1) gives, for every $r \in R$,

$$\|x - r\| = \varphi(Tq - r) = \varphi(Tq + w - r) = \|y - r\|.$$

Thus R does not resolve X . Since R was arbitrary, no finite resolving set exists. $\square$

Corollary 3. *Every polyhedral normed space of dimension at least two has infinite metric dimension. Every non-strictly-convex normed plane has infinite metric dimension.*

Proof. A full-dimensional polytope has a facet. In dimension two, every nondegenerate segment in the unit circle extends to a one-dimensional exposed face, which is a facet. $\square$

In particular, ℓ_1^d and ℓ_∞^d have infinite metric dimension. This also shows why strict convexity is indispensable in the planar version.

5. THE EXACT VALUE FOR ℓ_p^d

Theorem 4 (Coordinate-power spaces). *For $d \geq 1$ and $1 < p < \infty$,*

$$\text{mdim}(\ell_p^d) = d + 1.$$

More precisely, $\{0, e_1, \dots, e_d\}$ resolves ℓ_p^d . The same conclusion holds for every weighted norm

$$\|x\| = \left(\sum_{i=1}^d c_i |x_i|^p \right)^{1/p}, \quad c_i > 0,$$

and every invertible linear image of such a norm.

Proof. It suffices to treat the weighted form. Suppose $x, y \in \mathbb{R}^d$ have the same distances to $0, e_1, \dots, e_d$. Equality at the origin says

$$(2) \quad \sum_j c_j |x_j|^p = \sum_j c_j |y_j|^p.$$

Equality at e_i , after taking p th powers and subtracting (2), gives

$$|x_i - 1|^p - |x_i|^p = |y_i - 1|^p - |y_i|^p.$$

Set $h(t) = |t - 1|^p - |t|^p$ and $j_p(t) = \text{sgn}(t)|t|^{p-1}$. Then

$$h'(t) = p(j_p(t - 1) - j_p(t)) < 0,$$

because j_p is strictly increasing and $t - 1 < t$. Hence h is strictly decreasing, so $x_i = y_i$. This holds for every i , and therefore $x = y$. Theorem 1 proves optimality. An invertible linear isometry transports resolving sets and metric dimension. $\square$

6. A SMOOTH NON-HILBERT NEIGHBOURHOOD

The next result is useful because it does not rely on coordinate-power separability. All Euclidean norms and sufficiently small $C^{1,1}$ perturbations satisfy it.

Theorem 5 (Quantitative near-Hilbert class). *Let N be a norm on $\mathbb{R}^d$ and write*

$$\Phi(x) = N(x)^2 = |x|_2^2 + H(x).$$

Assume that H is continuously differentiable and that

$$|\nabla H(u) - \nabla H(v)|_2 \leq \eta |u - v|_2 \quad (u, v \in \mathbb{R}^d)$$

for some $\eta < 2/\sqrt{d}$. Then $\{0, e_1, \dots, e_d\}$ resolves $(\mathbb{R}^d, N)$, and consequently

$$\text{mdim}(\mathbb{R}^d, N) = d + 1.$$

Proof. Define the relative squared-distance map $G : \mathbb{R}^d \rightarrow \mathbb{R}^d$ by

$$G_i(x) = \Phi(x - e_i) - \Phi(x) = 1 - 2x_i + E_i(x), \quad E_i(x) = H(x - e_i) - H(x).$$

For $x, y \in \mathbb{R}^d$, the fundamental theorem of calculus on the segment from y to x yields

$$\begin{aligned} E_i(x) - E_i(y) &= \int_0^1 \langle \nabla H(y + t(x - y) - e_i) \\ &\quad - \nabla H(y + t(x - y)), x - y \rangle dt. \end{aligned}$$

The two gradient arguments in the integrand are one unit apart. Therefore

$$|E_i(x) - E_i(y)| \leq \eta |x - y|_2, \quad |E(x) - E(y)|_2 \leq \sqrt{d} \eta |x - y|_2.$$

If $G(x) = G(y)$, then

$$2|x - y|_2 = |E(x) - E(y)|_2 \leq \sqrt{d}\eta|x - y|_2.$$

Since $\sqrt{d}\eta < 2$, this forces $x = y$. Equal distances to the origin and the e_i imply equality of all the G_i , so the landmarks resolve. Apply Theorem 1 for optimality. $\square$

Corollary 6 (Explicit non-Hilbert examples). *For every $d \geq 2$, the norms*

$$N_\varepsilon(x) = (|x|_2^2 + \varepsilon\|x\|_4^2)^{1/2}$$

have metric dimension $d + 1$ for all sufficiently small $\varepsilon > 0$. They are smooth, strictly convex, and non-Hilbert.

Proof. The ℓ_2 -sum of two norms is a norm, and the Euclidean term makes it strictly convex. The function $x \mapsto \|x\|_4^2$ has globally Lipschitz gradient: away from zero its Hessian is homogeneous of degree zero and bounded on the Euclidean unit sphere, while the degree-one gradient extends Lipschitzly through zero. Hence Theorem 5 applies when ε is small enough. Finally, with $x = e_1$ and $y = e_2$, the two sides of the parallelogram identity contain respectively the unequal ℓ_4^2 contributions $2\varepsilon\sqrt{2}$ and 4ε . $\square$

7. WHY THE FULL STRICTLY CONVEX QUESTION IS NOT CLAIMED

Eight focused routes were checked. The successful ones are the lower bound, facet cone, coordinate-power separation, and quantitative near-Hilbert argument above. Four further routes identify genuine obstructions:

- Strictly convexifying a facet norm does not preserve its collisions: $\ell_q^d \rightarrow \ell_\infty^d$ as $q \rightarrow \infty$, yet every finite $q > 1$ has metric dimension $d + 1$.
- The topological obstruction ends exactly at d target coordinates; an extra distance coordinate removes the invariance-of-domain contradiction.
- Strict planar bisectors are direction-dependent graphs, but their strips are not uniform over the unknown pair. Incidence counting does not force a simplex avoiding every bisector.
- At infinity, smooth strictly convex norms encode direction through the dual Gauss map. Without a quantitative curvature bound, uniform asymptotic approximation does not prevent folds at finite or intermediate scale.

Thus the packet neither proves nor disproves $\text{mdim } X = d + 1$ for every strictly convex d -dimensional normed space.

8. VERIFICATION, LITERATURE CAUTION, AND REVIEW

All arguments are exact and analytic; no numerical evidence is used. The source was rebuilt from the cached arXiv file, and the displayed crop is taken from source PDF page 12. Cheap run indexes and targeted searches found no later primary-source answer, but terminology is unusually ambiguous and older work often says *determining set* or *equidistant set*. In particular, Kalisch–Straus [2] should be checked by a human specialist before any novelty claim. No priority claim is made.

Human review should focus on the intended parenthetical scope of Problem 1, the openness of the cone over a facet, and the global $C^{1,1}$ hypothesis in Theorem 5. The packet should not be cited as resolving the universal strictly convex case.

REFERENCES

- [1] G. P. Gehér, *Is it possible to determine a point lying in a simplex if we know the distances from the vertices?*, arXiv:1507.05114.
- [2] G. K. Kalisch and E. G. Straus, *On the determination of points in a Banach space by their distances from the points of a given set*, An. Acad. Brasil. Ci. **29** (1957), 501–519.